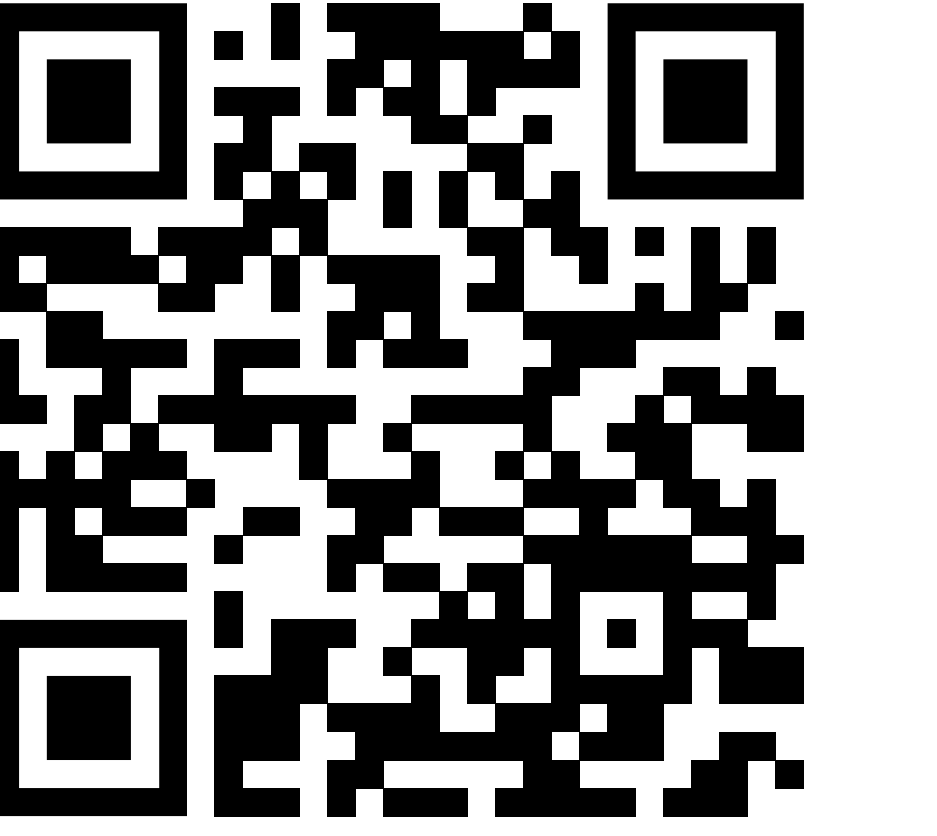
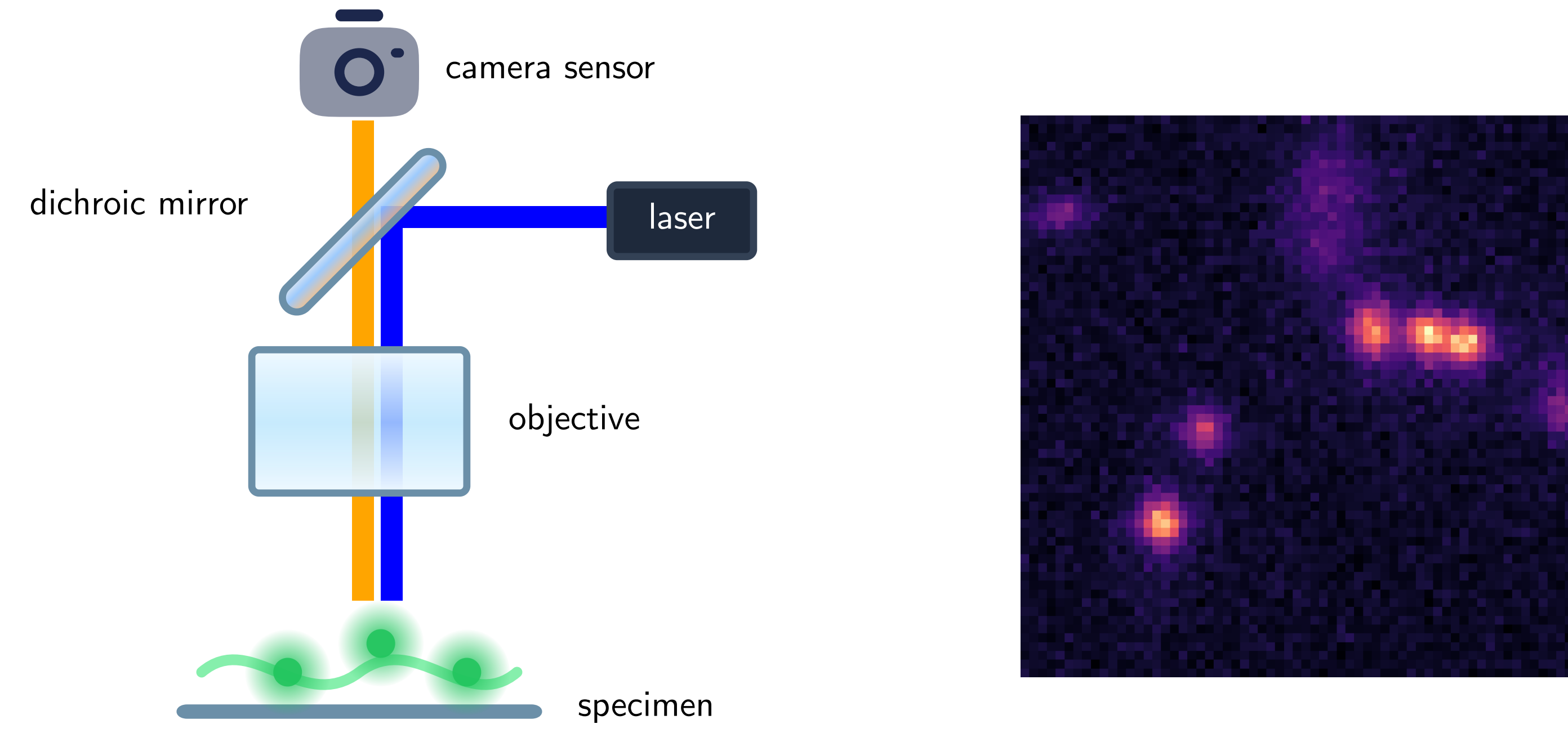


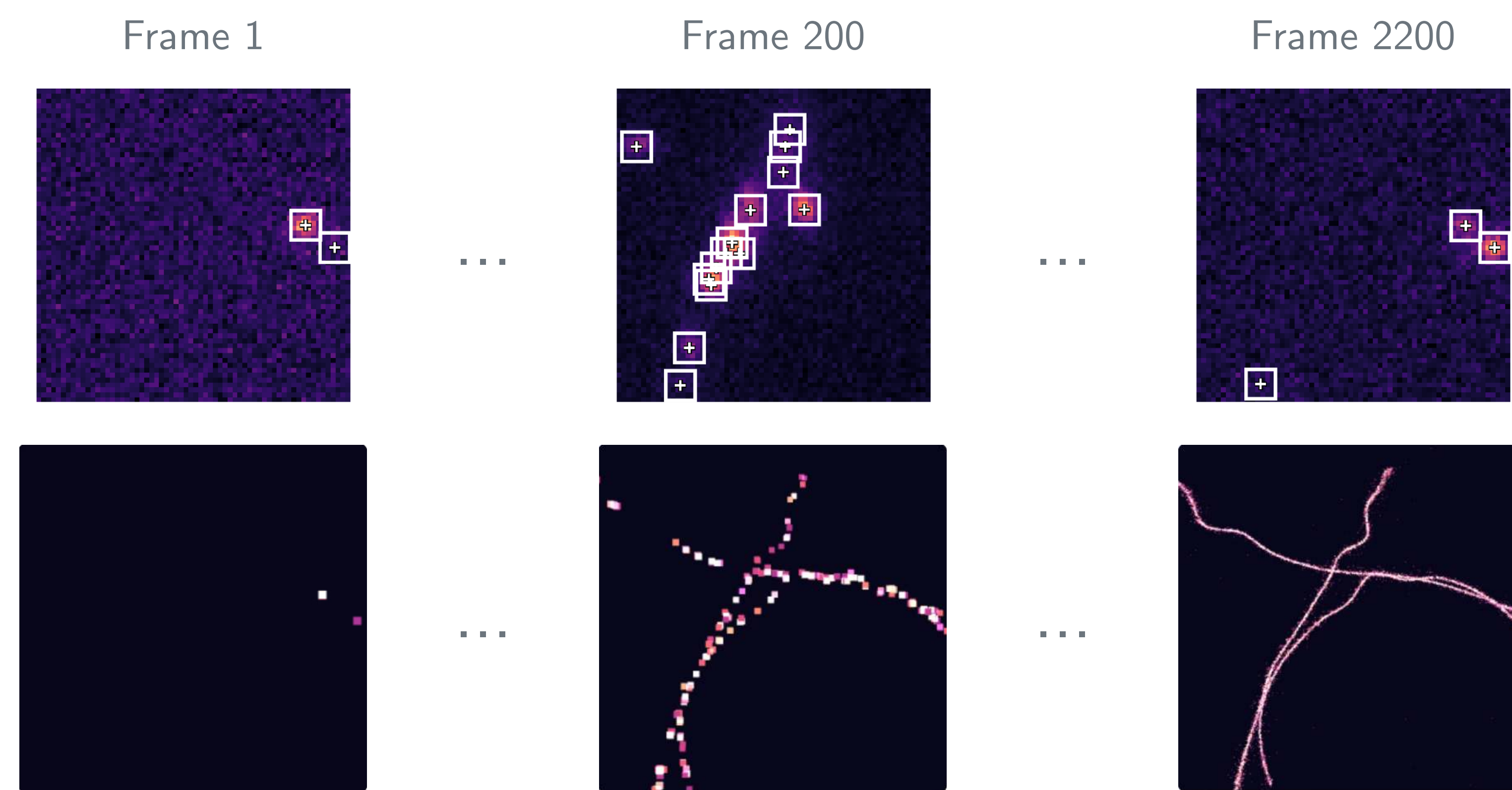


Single molecule localization microscopy

Light microscopy is intrinsically limited by the diffraction of light at around 200nm.



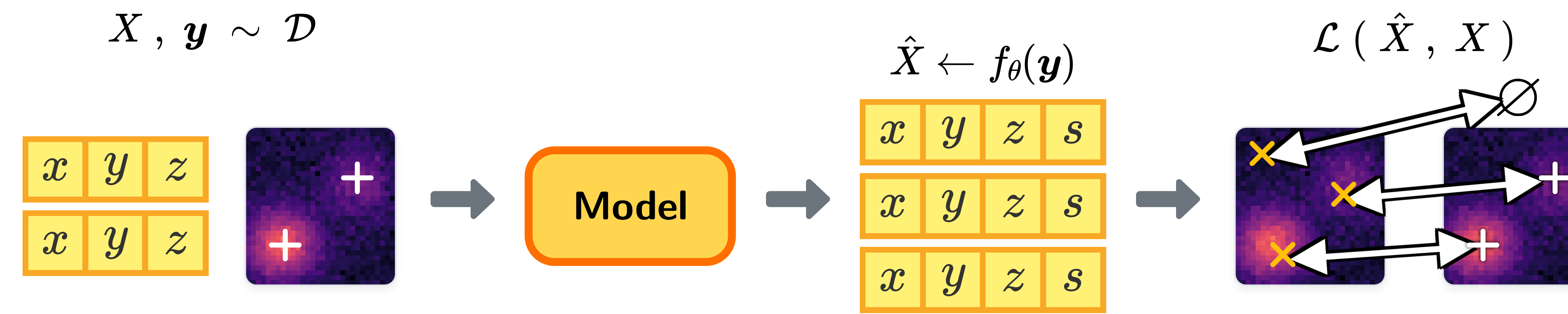
Single molecule localization microscopy (SMLM) detects and localizes individual fluorophores stained on the specimen to super-resolve acquisitions.



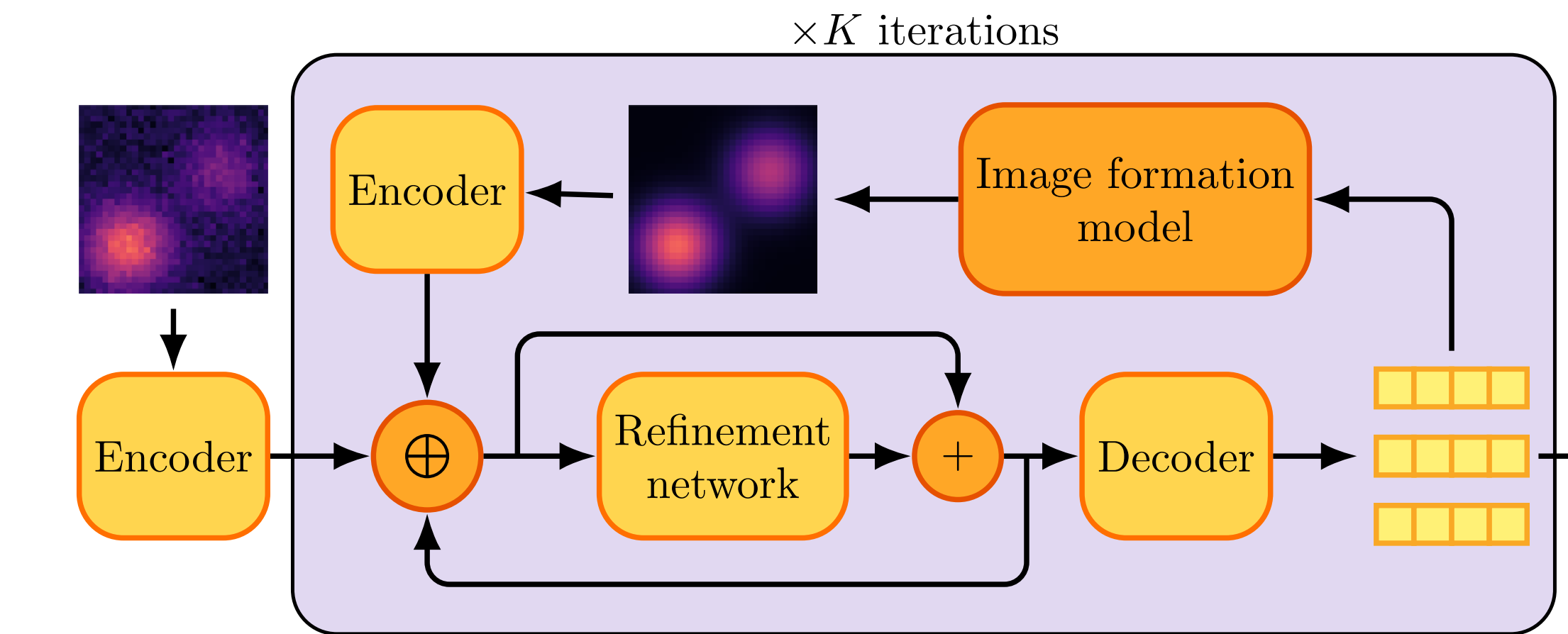
Can learning-based methods enable faster SMLM reconstruction?

Method

Following Speiser et al. (2021), we train our model using **supervised learning** on synthetic data generated by the simulator of Sage et al. (2019).



Inspired by the success of iterative refinement networks for inverse problems (Hur et al., 2019; Yu et al., 2023), we propose an **iterative architecture** that leverages the image formation process.



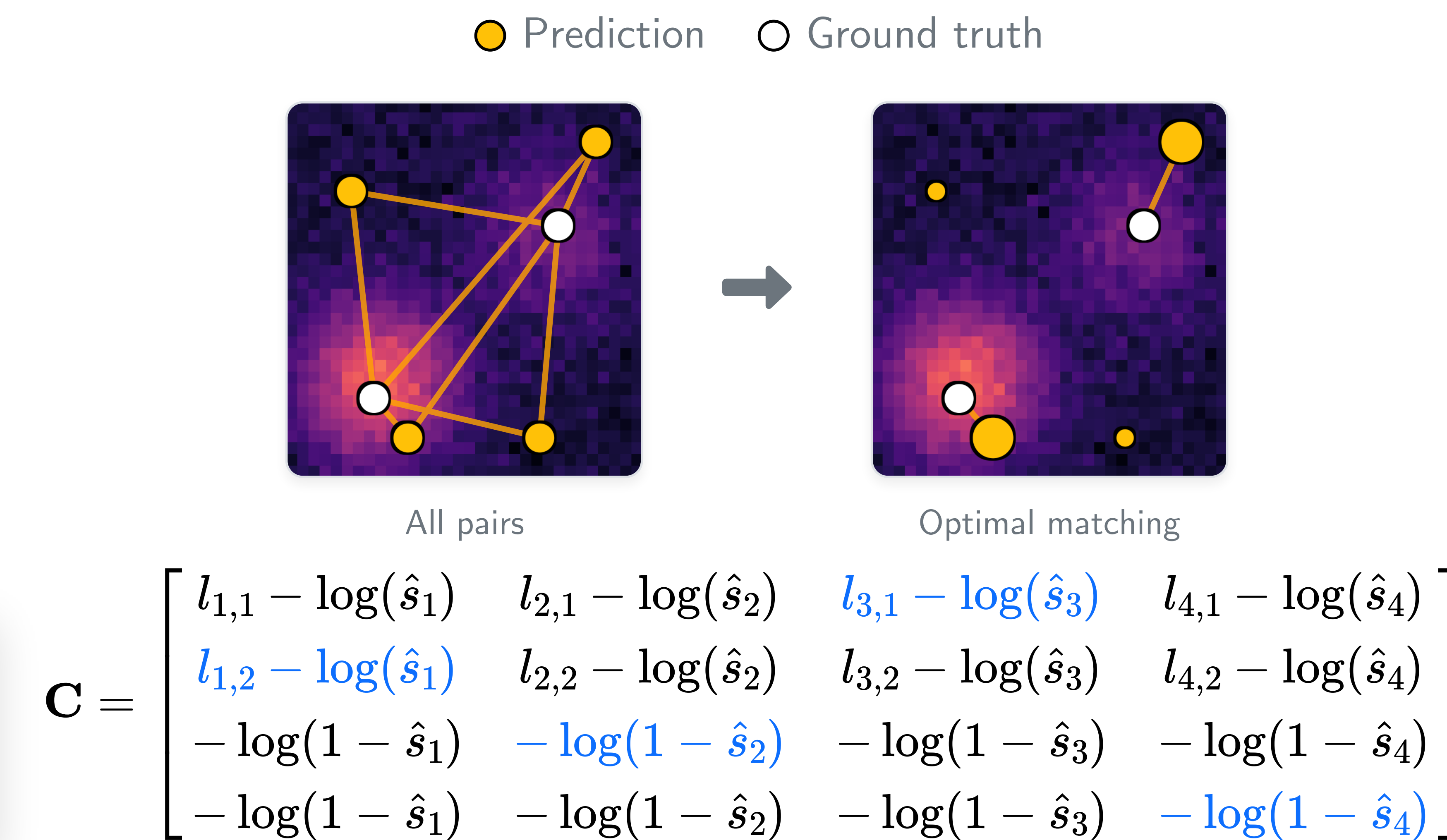
We also frame SMLM as a set matching problem with an **optimal-transport-based loss function**.

- Compute pairwise localization costs:

$$l_{i,j} = (\hat{\mathbf{x}}_i - \mathbf{x}_j)^\top \Sigma^{-1} (\hat{\mathbf{x}}_i - \mathbf{x}_j) + \log \det(\Sigma)$$
- Build a square cost matrix $\mathbf{C}$ combining localization cost and detection penalty (binary cross-entropy).
- Compute the optimal transport plan Γ^* and use the min. cost as the loss function:

$$\mathcal{L}(\hat{X}, X) = \langle \Gamma^* | \mathbf{C} \rangle_{\mathcal{F}}$$

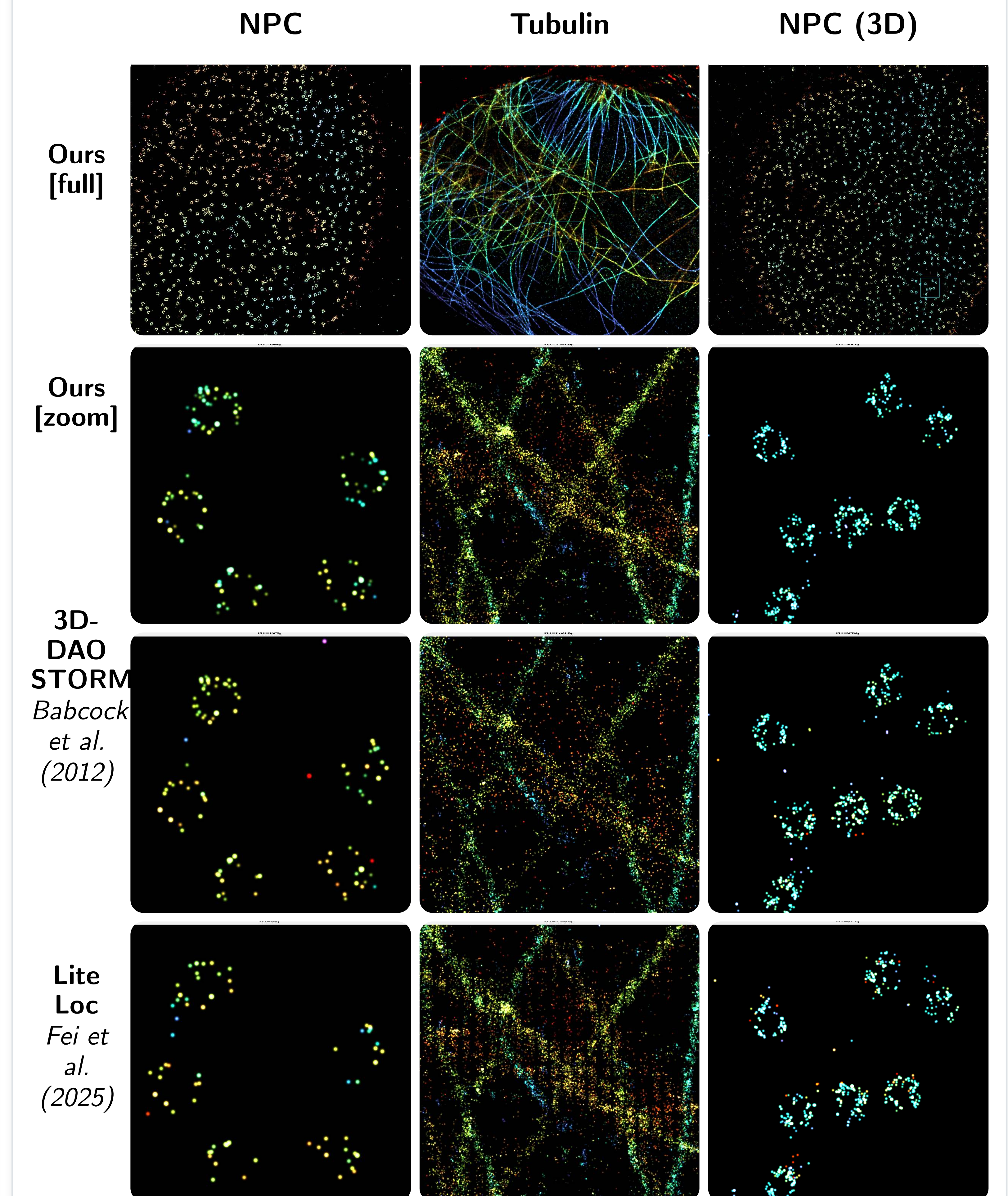
where $\Gamma^* = \arg \min_{\Gamma \in \mathbb{B}} \langle \Gamma | \mathbf{C} \rangle_{\mathcal{F}} - \epsilon H(\Gamma)$



Results

Our method achieves **better quantitative results** on both synthetic and real benchmarks (see paper).

Qualitative comparisons on experimental data:



Limitation: Our method relies on **synthetic data**; future work could explore blind SMLM setups.